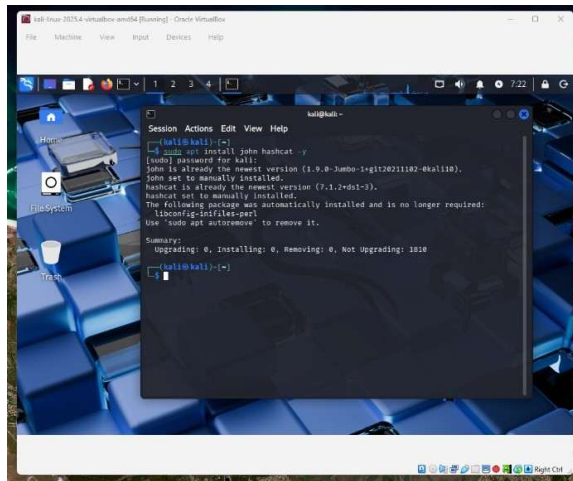


Practical 03

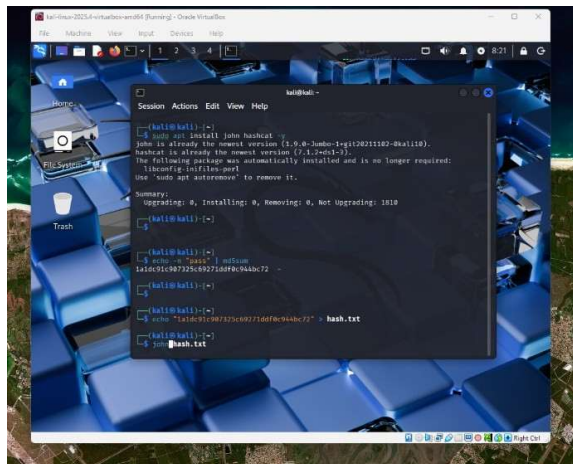
Important: This practical must be performed only on sample/test data in a controlled environment. Do not attempt password cracking on real systems or unauthorized data.

1. Open Terminal.
2. Install Tools (if needed). Run the following command
 - a. `sudo apt install john hashcat -y`



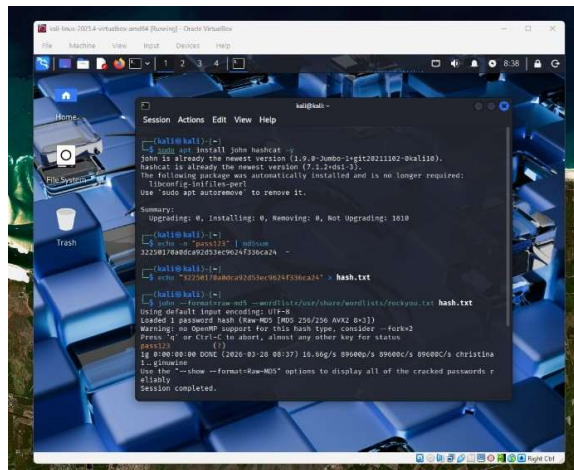
```
kali@kali:~$ sudo apt install john hashcat -y
[sudo] password for kali:
john is already the newest version (1.9.0-Jumbo-1-g174021182-4kali10).
john set to manually installed.
hashcat is already the newest version (7.1.2-6kali-3).
hashcat set to manually installed.
The following package was automatically installed and is no longer required:
  libconfig-inifiles-perl
Use 'sudo apt autoremove' to remove it.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1010
kali@kali:~$
```

3. Generate a Sample Password Hash. Run:
 - a. `echo -n "pass123" | md5sum`
Note: The -n option avoids adding a newline character, ensuring correct hash generation.
 - b. The output will display: `1a1d....`



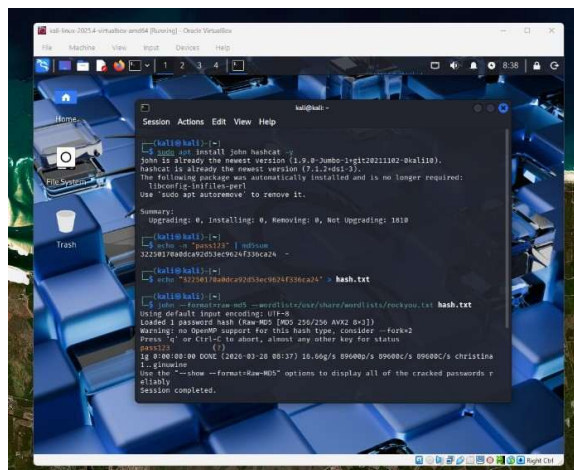
```
kali@kali:~$ echo -n "pass123" | md5sum
1a1dc91c9b7225c9271d0f9c944bc72 -
kali@kali:~$ echo "1a1dc91c9b7225c9271d0f9c944bc72" > hash.txt
kali@kali:~$
```

4. Create Hash File. Run the following command:
 - a. `echo "1a1d...." > hash.txt`



5. Crack Password using John the Ripper

a. john hash.txt



Note: If cracking takes too long, use a wordlist-based approach for faster results.

6. Output should show:

a. pass123

